the happy path: on human agency and AI interfaces

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the problem. it's been said that "I want AI to do my laundry and dishes so that I can do art and writing, not for AI to do my art and writing so that I can do my laundry and dishes" (first to my knowledge shared by Joanna Maciejewska). an increasingly likely future is one in which AI does both and we do nothing

lots of research has been dedicated to the question of how AI can make humans more efficient. but, if agency is the freedom and ability to change the world by acting on it, in an increasingly automated world, how do we ensure people can express their agency in ways they find most compelling? in a world where AI could do anything we can (except where being human is the point, like sports), how can we make sure AI values human agency? we can't assume that the default is for future advanced AI to collaborate with us like humans do, or even at all

the past. many have noted that an analogous question was raised after computers beat humans in chess. it has now been decades since an unassisted human has defeated a state of the art chess engine. and yet, chess has actually since experienced a resurgence, and chess players are better than ever at all levels

i distinctly remember prof. noah goodman (my phd advisor) making this comparison at a neurips workshop years ago when asked on a panel about the future of math: it could become increasingly like chess, where experts would come to terms with the reality that they could not produce math as advanced as the top AI algorithms. in this world, mathematicians would study math not because they are necessary for these advances, but because they find fulfillment in its pursuit

increasingly, this seems likely, not only for chess for a growing list of human passions. there is a chance we may even find ourselves temporarily in a world where people can find no way to contribute. however, there is also a potential world where we may be able to benefit from advanced AI without sacrificing all of our agency. for this, AI must actively find ways to people to meaningfully contribute if they would like. so, we must design AI to value and prioritize human agency

the bare minimum. in the same way that companies may hire interns or professors take on students where the cost of mentorship is nontrivial, agency must be fundamental

first, people should have the opportunity to automate tasks that they would prefer not to do. this has already happened in many professions. today, chairs and tables are made mostly by machines - still, woodworking is still a thriving human pursuit, as are sewing and printing. in a world where more tasks can be automated, the lowest bar is to make sure that people still have opportunities to pursue even tasks that can theoretically be automated. my point here isn't to slow progress but that we should be able to choose where to "exert" agency and be sure that any sacrifice of agency is an explicit choice

the hope. but, allowing people to pursue their passions is the lowest bar. we can aim higher. future AI systems should not simply give people tasks that make them feel useful, like asking a child to blow on a finished painting to help it dry or finding the perfect spot to plant a flower. agency is inherently a source of fulfillment for people, but false agency can be worse than none at all: we shouldn't settle for the feeling of usefulness and impact, when we have none

that is: existing human-AI collaboration schemes are early and quite naive – but, future agentic systems (AI agents acting together) will be incredibly effective task allocators. if someone sincerely wants to contribute or learn, a good manager/mentor will try to find a way to let them (even if resources are needed to catch them up to speed). planned well, if someone applies their efforts wholeheartedly, the resulting project will most likely be better than if they had not participated

only for tasks where people fundamentally cannot help should we fall back to the lower bar of allowing humans to pursue them if they would like, but even in these cases there may be ways to incorporate these efforts into some larger project. still, for many problems, especially those limited by interaction with the real world, humans will be not-meaningfully-worse than even incredibly efficient agents

the future. to reiterate, we should not expect our agentic systems to have this property by default. but, this requires that we guarantee agency is at their core. in this world, AI could be collaborators, trusted advisors, attentive listeners, and caring aids. those who want to abandon their agency should have that freedom; but, most would quickly get bored in a world where they don't do or choose anything. on the other side, AI that values our agency would help us discover and explore our passions. i have at times been very fortunate to have advisors both vastly more experienced than myself while also being thoughtful, patient, caring communicators – i am optimistic for a world where everyone has this experience

there is a future where we collaborate with advanced AI, meaningfully contributing and learning. in UX, there is an idea of a happy path: a way of working with software that aligns with its design and goes smoothly. for the best chance of a happy path with powerful agentic systems, we should (1: AI-human) ensure agentic systems value human agency and (2: human-AI) design frictionless interfaces, allowing people effortless access to AI

for future AI systems to empower us, prioritize agency from the start